Chapter 2: Basic Topology

Section: Metric Spaces

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Metric Spaces: Definitions and Setup

Topological Concepts

Open and Closed Sets

Closure and Relative Topology

Metric Spaces: Definitions and Setup

Definition 2.15: Metric Space

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A set X is a metric space if with any two points $p, q \in X$ there is associated a real number $d(p, q)$, called the distance, such that:

(a) $d(p, q) > 0$ if $p \neq q$; $\quad d(p, p) = 0$

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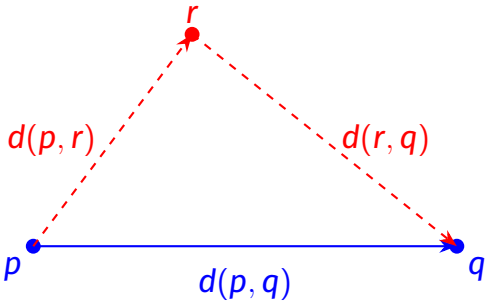
(b) $d(p, q) = d(q, p)$

(c) $d(p, q) \leq d(p, r) + d(r, q)$, for any $r \in X$

Any function with these three properties is called a *distance function* or *metric*.

Definition 2.15: Visualizing the Triangle Inequality



$$d(p,q) \leq d(p,r) + d(r,q)$$

Examples 2.16: Euclidean Spaces

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The most important metric spaces are the Euclidean spaces R^k:

$$d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}| \qquad (\mathbf{x}, \mathbf{y} \in R^k)$$

Special cases: R^1 (the real line), R^2 (the complex plane).

By Theorem 1.37, this satisfies Definition 2.15.

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Special cases: R^1 (the real line), R^2 (the complex plane).

By Theorem 1.37, this satisfies Definition 2.15.

Key observation: Every subset Y of a metric space X is itself a metric space (with the same distance function).

Definition 2.17: Segments and Intervals

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- Segment (a, b): all real x with $a < x < b$ (open)
- Interval $[a, b]$: all real x with $a \leq x \leq b$ (closed)

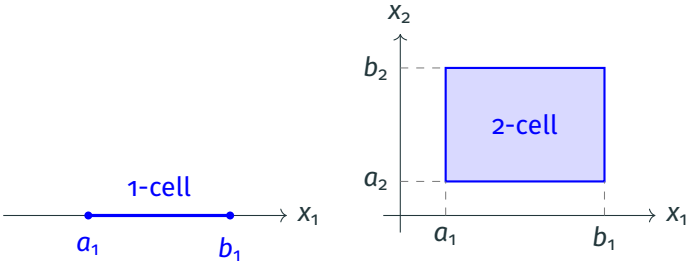
Also: half-open intervals $[a, b)$ and $(a, b]$.

Definition 2.17: Segments and Intervals

Definition 2.17: k-cells

If $a_i < b_i$ for $i = 1, \ldots, k$, the set of all $\mathbf{x} = (x_1, \ldots, x_k) \in R^k$ with $a_i \leq x_i \leq b_i$ for $1 \leq i \leq k$ is a k-cell.

A 1-cell is an interval. A 2-cell is a rectangle.



Definition 2.17: Open and Closed Balls

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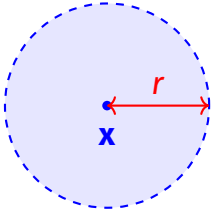
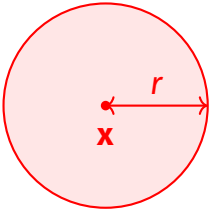
For $\mathbf{x} \in R^k$ and $r > 0$:

- Open ball: $B(\mathbf{x}, r) = \{\mathbf{y} \in R^k : |\mathbf{y} - \mathbf{x}| < r\}$
- Closed ball: $\bar{B}(\mathbf{x}, r) = \{\mathbf{y} \in R^k : |\mathbf{y} - \mathbf{x}| \leq r\}$



Open ball: boundary *not* included



Closed ball: boundary included

Definition 2.17: Convex Sets

Definition 2.17: Convexity

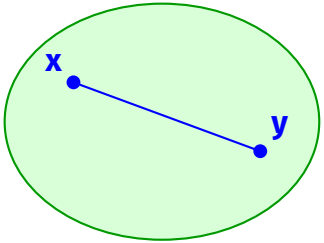
A set $E \subset R^k$ is convex if

$$\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in E$$

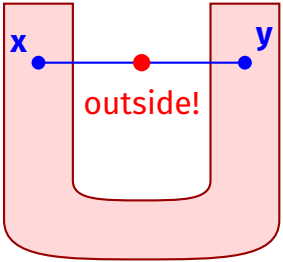
whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$.

Fact: Balls are convex. k-cells are convex.

Definition 2.17: Convex vs. Non-Convex Sets



Convex



Not convex

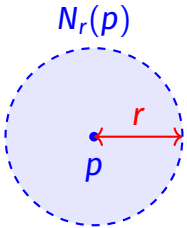
Topological Concepts

Definition 2.18: Topological Vocabulary

Let X be a metric space. All points and sets are elements and subsets of X.

Definition 2.18(a): Neighborhood

A neighborhood of p is a set $N_r(p)$ consisting of all q such that $d(p, q) < r$, for some $r > 0$.

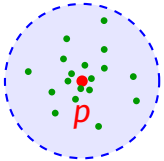


Definition 2.18: Topological Vocabulary

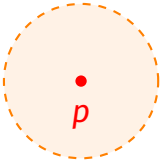
Definition 2.18(b)–(c): Limit Points and Isolated Points

(b) p is a limit point of E if *every* neighborhood of p contains a point $q \neq p$ with $q \in E$.

(c) If $p \in E$ and p is *not* a limit point of E, then p is an isolated point of E.



Limit point: points of E crowd around p



Isolated point: no other points of E nearby

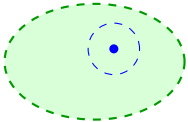
Definition 2.18: Topological Vocabulary

Definition 2.18(d)–(f): Closed, Interior, Open

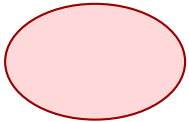
(d) E is closed if every limit point of E is a point of E.

(e) p is an interior point of E if there is a neighborhood N of p with $N \subset E$.

(f) E is open if every point of E is an interior point of E.



Open: room around every point



Closed: contains all limit points

Definition 2.18: Topological Vocabulary

Definition 2.18(g)–(h): Complement and Perfect

(g) The complement of E is $E^c = \{p \in X : p \notin E\}$.

(h) E is perfect if E is closed and every point of E is a limit point of E.

Perfect $=$ closed $+$ no isolated points.

Definition 2.18: Topological Vocabulary

Definition 2.18(i)–(j): Bounded and Dense

(i) E is bounded if there exist $M > 0$ and $q \in X$ such that $d(p, q) < M$ for all $p \in E$.

(j) E is dense in X if every point of X is a limit point of E, or a point of E (or both).

Note: In R^1 neighborhoods are segments; in R^2 neighborhoods are open disks.

Theorem 2.19: Every Neighborhood Is an Open Set

Theorem 2.19

Every neighborhood is an open set.

Proof of Theorem 2.19

Proof

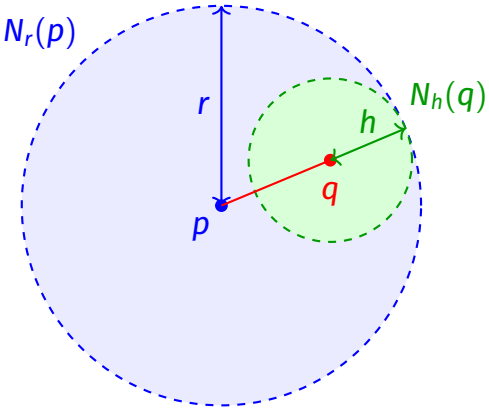
Let $E = N_r(p)$ and let $q \in E$. Set $h = r - d(p, q) > 0$.

For any s with $d(q, s) < h$:

$$d(p, s) \leq d(p, q) + d(q, s) < (r - h) + h = r$$

So $s \in E$. Thus $N_h(q) \subset E$, and q is an interior point. $\square$

Proof of Theorem 2.19 — Diagram



Theorem 2.20: Limit Points Yield Infinitely Many Points

Theorem 2.20

If p is a limit point of E, then every neighborhood of p contains infinitely many points of E.

Proof of Theorem 2.20

Proof (by contradiction)

Suppose some neighborhood N of p contains only finitely many points of E (other than p): call them $q_1, \ldots, q_n$.

Let $r = \min_{1 \leq m \leq n} d(p, q_m) > 0$.

Then $N_r(p)$ contains *no* point of E other than $p \implies p$ is not a limit point. Contradiction! $\square$

Corollary: A finite point set has no limit points.

Examples 2.21: Subsets of R^2

Examples 2.21

Consider the following subsets:

(a) $\{z \in \mathbb{C} : |z| < 1\}$ — open unit disk

(b) $\{z \in \mathbb{C} : |z| \leq 1\}$ — closed unit disk

(c) A nonempty finite set

Examples 2.21: Subsets of R^2

Examples 2.21 (continued)

(d) The set of all integers $\mathbb{Z}$

(e) $\{1/n : n = 1, 2, 3, \dots\}$

(f) $\mathbb{C}$ (all complex numbers, i.e., R^2)

(g) The segment (a, b)

Note on (e): E has a limit point ($z = 0$) that is *not in* E.

Examples 2.21: Subsets of R^2

Properties Table

	Closed	*Open*	*Perfect*	*Bounded*
(a) open disk	No	Yes	No	Yes
(b) closed disk	Yes	No	Yes	Yes
(c) finite set	Yes	No	No	Yes
(d) $\mathbb{Z}$	Yes	No	No	No
(e) $\{1/n\}$	No	No	No	Yes
(f) $\mathbb{C}$	Yes	Yes	Yes	No
(g) (a, b)	No	$*$	No	Yes

$*$ Open in R^1 but not in R^2!

Open and Closed Sets

Theorem 2.22: De Morgan's Law for Sets

Theorem 2.22

For any collection $\{E_\alpha\}$ of sets:

$$\left(\bigcup_\alpha E_\alpha\right)^c = \bigcap_\alpha (E_\alpha^c)$$

Proof of Theorem 2.22 — $A \subset B$

Proof: $A \subset B$

Let $A = (\bigcup_\alpha E_\alpha)^c$ and $B = \bigcap_\alpha (E_\alpha^c)$.

$x \in A \Rightarrow x \notin \bigcup_\alpha E_\alpha \Rightarrow x \notin E_\alpha$ for all $\alpha \Rightarrow x \in E_\alpha^c$ for all $\alpha \Rightarrow x \in B$.

Proof of Theorem 2.22 — $B \subset A$

Proof

Let $A = (\bigcup_\alpha E_\alpha)^c$ and $B = \bigcap_\alpha (E_\alpha^c)$.

$x \in A \Rightarrow x \notin \bigcup_\alpha E_\alpha \Rightarrow x \notin E_\alpha$ for all $\alpha \Rightarrow x \in E_\alpha^c$ for all $\alpha \Rightarrow x \in B$.

$x \in B \Rightarrow x \in E_\alpha^c$ for all $\alpha \Rightarrow x \notin E_\alpha$ for all $\alpha \Rightarrow x \notin \bigcup_\alpha E_\alpha \Rightarrow x \in A$.

Hence $A = B$. $\square$

Theorem 2.23: Open Sets and Closed Complements

Theorem 2.23

A set E is open if and only if its complement E^c is closed.

Corollary: A set F is closed if and only if F^c is open.

Proof of Theorem 2.23 — Forward Direction

E^c closed $\Longrightarrow$ E open

Suppose E^c is closed. Pick $x \in E$.

Then $x \notin E^c$ and x is not a limit point of E^c.

Proof of Theorem 2.23 — Forward Direction

E^c closed $\Longrightarrow$ E open

Suppose E^c is closed. Pick $x \in E$.

Then $x \notin E^c$ and x is not a limit point of E^c.

So there exists a neighborhood N of x with $E^c \cap N = \emptyset$.

This means $N \subset E$, so x is an interior point.

Proof of Theorem 2.23 — Forward Direction

E^c closed $\Longrightarrow$ E open

Suppose E^c is closed. Pick $x \in E$.

Then $x \notin E^c$ and x is not a limit point of E^c.

So there exists a neighborhood N of x with $E^c \cap N = \emptyset$.

This means $N \subset E$, so x is an interior point.

Since x was arbitrary, E is open. $\square$

Proof of Theorem 2.23 — Reverse Direction

E open $\implies E^c$ closed

Suppose E is open. Let x be a limit point of E^c.

Every neighborhood of x contains a point of E^c.

So x is *not* an interior point of E.

Proof of Theorem 2.23 — Reverse Direction

E open $\Longrightarrow$ E^c closed

Suppose E is open. Let x be a limit point of E^c.

Every neighborhood of x contains a point of E^c.

So x is *not* an interior point of E.

Since E is open, $x \notin E$. Hence $x \in E^c$.

All limit points of E^c lie in E^c, so E^c is closed. $\square$

Theorem 2.24: Unions and Intersections

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(a) For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.

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(c) For any finite collection $G_1, \ldots, G_n$ of open sets, $\bigcap_{i=1}^n G_i$ is open.

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(d) For any finite collection $F_1, \ldots, F_n$ of closed sets, $\bigcup_{i=1}^n F_i$ is closed.

Proof of Theorem 2.24(a)

Proof of (a): Arbitrary union of open sets is open

Let $G = \bigcup_\alpha G_\alpha$.

If $x \in G$, then $x \in G_\alpha$ for some α.

Since G_α is open, x is an interior point of G_α.

Proof of Theorem 2.24(a)

Proof of (a): Arbitrary union of open sets is open

Let $G = \bigcup_\alpha G_\alpha$.

If $x \in G$, then $x \in G_\alpha$ for some α.

Since G_α is open, x is an interior point of G_α.

So there is a neighborhood $N \subset G_\alpha \subset G$.

Hence x is an interior point of G, and G is open. $\square$

Proof of Theorem 2.24(c)

Proof of (c): Finite intersection of open sets is open

Let $H = \bigcap_{i=1}^{n} G_i$. For $x \in H$:

Each G_i is open, so there exist neighborhoods N_i of x with radius r_i such that $N_i \subset G_i$.

Proof of Theorem 2.24(c)

Proof of (c): Finite intersection of open sets is open

Let $H = \bigcap_{i=1}^{n} G_i$. For $x \in H$:

Each G_i is open, so there exist neighborhoods N_i of x with radius r_i such that $N_i \subset G_i$.

Set $r = \min(r_1, \ldots, r_n) > 0$ (finite minimum of positive numbers).

Proof of Theorem 2.24(c)

Proof of (c): Finite intersection of open sets is open

Let $H = \bigcap_{i=1}^{n} G_i$. For $x \in H$:

Each G_i is open, so there exist neighborhoods N_i of x with radius r_i such that $N_i \subset G_i$.

Set $r = \min(r_1, \ldots, r_n) > 0$ (finite minimum of positive numbers).

Then $N_r(x) \subset G_i$ for all i, so $N_r(x) \subset H$. $\square$

Examples 2.25: Why Finiteness Matters

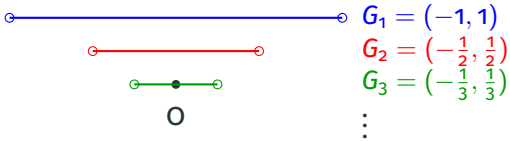
Example 2.25

Let $G_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$ for $n = 1, 2, 3, \ldots$

Each G_n is open. But:

$$\bigcap_{n=1}^{\infty} G_n = \{0\}$$

which is not open.



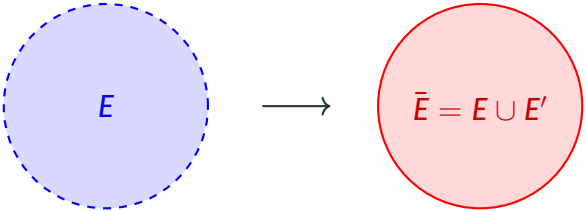
Closure and Relative Topology

Definition 2.26: Closure of a Set

Definition 2.26

If $E \subset X$ and E' denotes the set of all limit points of E in X, then the closure of E is:

$$\bar{E} = E \cup E'$$



Theorem 2.27: Properties of Closure

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If X is a metric space and $E \subset X$, then:

(a) $\bar{E}$ is closed.

Theorem 2.27: Properties of Closure

Theorem 2.27

If X is a metric space and $E \subset X$, then:

(a) $\bar{E}$ is closed.

(b) $E = \bar{E}$ if and only if E is closed.

Theorem 2.27: Properties of Closure

Theorem 2.27

If X is a metric space and $E \subset X$, then:

(a) $\bar{E}$ is closed.

(b) $E = \bar{E}$ if and only if E is closed.

(c) $\bar{E} \subset F$ for every closed set $F \subset X$ with $E \subset F$.

By (a) and (c): $\bar{E}$ is the smallest closed set containing E.

Proof of Theorem 2.27

Proof of (a): $\bar{E}$ is closed

If $p \notin \bar{E}$, then $p \notin E$ and $p \notin E'$.

So p has a neighborhood N that does not intersect E.

Hence the complement of $\bar{E}$ is open, so $\bar{E}$ is closed. $\square$

Proof of Theorem 2.27

Proof of (b): $E = \bar{E}$ iff E is closed

($\Rightarrow$) If $E = \bar{E}$, then E is closed by (a).

($\Leftarrow$) If E is closed, then $E' \subset E$, so $\bar{E} = E \cup E' = E$. $\square$

Proof of Theorem 2.27

Proof of (c): $\bar{E} \subset F$ for closed $F \supset E$

If F is closed and $F \supset E$:

$$E \subset F \Rightarrow E' \subset F' \quad \text{and} \quad F \text{ closed} \Rightarrow F' \subset F$$

So $E' \subset F' \subset F$, hence $F \supset E \cup E' = \bar{E}$. $\square$

Theorem 2.28: Supremum Belongs to the Closure

Theorem 2.28

Let E be a nonempty set of real numbers bounded above, and let $y = \sup E$. Then $y \in \bar{E}$.

Hence $y \in E$ if E is closed.

Proof of Theorem 2.28

Proof

If $y \in E$, then $y \in \bar{E}$.

Suppose $y \notin E$. For every $h > 0$, there exists $x \in E$ with

$$y - h < x < y$$

Proof of Theorem 2.28

Proof

If $y \in E$, then $y \in \bar{E}$.

Suppose $y \notin E$. For every $h > 0$, there exists $x \in E$ with

$$y - h < x < y$$

(otherwise $y - h$ would be an upper bound of E, contradicting $y = \sup E$).

Proof of Theorem 2.28

Proof

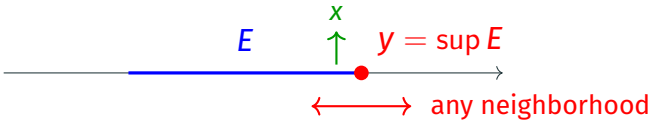
If $y \in E$, then $y \in \bar{E}$.

Suppose $y \notin E$. For every $h > 0$, there exists $x \in E$ with

$$y - h < x < y$$

(otherwise $y - h$ would be an upper bound of E, contradicting $y = \sup E$).

So every neighborhood of y contains a point of $E \Rightarrow y \in E' \Rightarrow y \in \bar{E}$. $\square$



Remark 2.29: Open Relative to a Subspace

Remark 2.29

Suppose $E \subset Y \subset X$.

E is open relative to Y if: for each $p \in E$, there exists $r > 0$ such that

$$q \in E \quad \text{whenever} \quad d(p,q) < r \ \text{ and } \ q \in Y.$$

Recall: (a,b) is open in R^1 but not open in R^2.

Theorem 2.30: Characterizing Relatively Open Sets

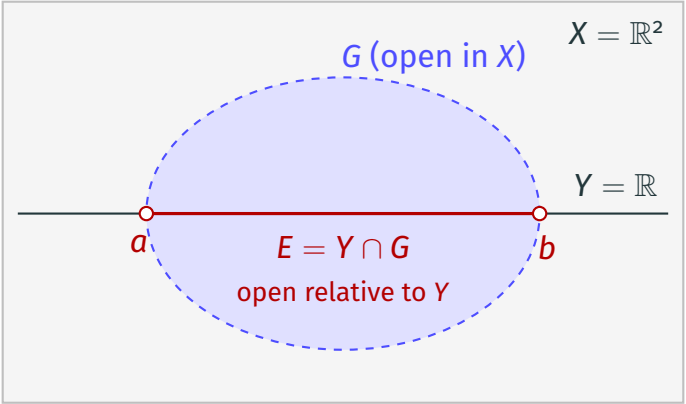
Theorem 2.30

Suppose $Y \subset X$. A subset E of Y is open relative to Y if and only if

$$E = Y \cap G$$

for some open subset G of X.

Theorem 2.30: Illustration



Proof of Theorem 2.30 — Forward Direction

E open relative to $Y \implies E = Y \cap G$

For each $p \in E$, choose $r_p > 0$ so that $q \in E$ whenever $d(p, q) < r_p$ and $q \in Y$.

Let $V_p = \{q \in X : d(p, q) < r_p\}$ and set $G = \bigcup_{p \in E} V_p$.

G is open in X (union of neighborhoods, by Theorems 2.19 and 2.24).

Proof of Theorem 2.30 — Forward Direction

E open relative to $Y \implies E = Y \cap G$

For each $p \in E$, choose $r_p > 0$ so that $q \in E$ whenever $d(p, q) < r_p$ and $q \in Y$.

Let $V_p = \{q \in X : d(p, q) < r_p\}$ and set $G = \bigcup_{p \in E} V_p$.

G is open in X (union of neighborhoods, by Theorems 2.19 and 2.24).

Claim: $E = G \cap Y$.

- $E \subset G \cap Y$: each $p \in E$ lies in $V_p \subset G$, and $p \in Y$.
- $G \cap Y \subset E$: if $q \in V_p \cap Y$, then $q \in E$ by choice of r_p.

$\square$

Proof of Theorem 2.30 — Converse Direction

$E = Y \cap G$ with G open $\Longrightarrow$ E open relative to Y

If G is open in X and $E = G \cap Y$:

For every $p \in E$, since $p \in G$ and G is open, there exists a neighborhood $V_p \subset G$.

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Then $V_p \cap Y \subset G \cap Y = E$.

So E is open relative to Y. $\square$

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- Open and closed are *dual*: E is open iff E^c is closed (Thm. 2.23). Unions of open sets are open; intersections of closed sets are closed. Finiteness is needed for the reverse (Thm. 2.24).

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- Open and closed are *dual*: E is open iff E^c is closed (Thm. 2.23). Unions of open sets are open; intersections of closed sets are closed. Finiteness is needed for the reverse (Thm. 2.24).
- The closure $\bar{E} = E \cup E'$ is the smallest closed set containing E (Thm. 2.27). Relatively open sets are characterized by $E = Y \cap G$ (Thm. 2.30).

Coming Up Next

Coming up next: Compact sets — a powerful finiteness condition.